

Cost-Aware Evaluation of Multi-Agent LLM Systems for Thai Sarcasm Detection

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github.com/Thanaphumi2006/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection

ABSTRACT. Multi-agent LLM architectures, verifier pipelines, debate panels, hybrid committees, are widely assumed to justify their extra inference cost with higher quality. We test that assumption on a task where surface form and intent diverge sharply: sarcasm detection in Thai (ประชด). On a 127-item human-labeled gold set drawn from product reviews and tweets, we compare five systems under identical measurement code, tracking F1, API calls, dollar cost, and latency simultaneously. A two-agent screener-verifier pipeline initially appears to win (F1 0.744 vs. 0.690). The advantage does not survive a fair baseline: letting the single agent read its own token-level confidence and tuning one threshold, chosen leave-one-fold-out, at zero added cost, lifts it to 0.725, after which no multi-agent system separates from it (best Δ F1 +0.019, 95% CI $[-0.073, +0.113]$, paired bootstrap). Model choice dominates architecture choice: a model 6 $\times$ cheaper than the flagship stays within a 0.05 F1 band, making a cheap single agent statistically indistinguishable from a flagship pipeline at one-eleventh the cost. We further establish a \$0 floor, three independent free methods (a 3-character regex, an embedding kNN, and a fine-tuned Thai transformer) all converge at F1 $\approx$ 0.59, and show a three-way confidence router that retains 95% of flagship quality at 40% of the cost. In the deployed cascade the pattern recurs once more: a free abstention threshold on the lexical tier (F1 0.627 $\rightarrow$ 0.806 at 42% of the calls) beats inserting a trained model tier, which a constant-output control ties exactly. Expanding the gold set 4.7 $\times$ (595 items, 104 positive) with adversarially plausible negatives narrows the pipeline-vs-tuned-baseline confidence interval from 0.186 to 0.056 wide and preserves the null result, with the point estimate now favouring the single agent (-0.019 , 91% chance the pipeline is not better) at 2.96 $\times$ the cost. At that scale the confidence-reading upgrade itself becomes statistically significant for the first time ($+0.040$, 95% CI $[+0.018, +0.060]$), while precision collapses for every architecture equally: hard negatives are a property of the data that additional agents do not repair. On a uniformly random test set (4.9% sarcastic, the true base rate versus 17--24% in the mined sets), every headline number collapses: the free cue detector falls from F1 0.59 to 0.09 and argmax GPT flags 36% of all text (F1 0.20), while the free confidence-reading upgrade is what rescues the system to F1 0.46, its largest single gain anywhere. We conclude that for subjective, low-resource classification, the first questions are not architectural: exhaust the information one agent already returns, then choose the cheapest sufficient model.

1 Introduction

Multi-agent LLM systems promise quality through redundancy: multiple calls, multiple roles, a final adjudication. The promise is easy to believe and expensive to hold, every added agent multiplies API cost and latency. Whether the quality materializes is an empirical question, and one that is rarely asked with the cost axis in view.

We ask it on Thai sarcasm detection. Sarcasm (ประชด) is a deliberately adversarial target: the writer states the opposite of their intent, so surface cues mislead exactly the models that rely on them. Thai adds a low-resource setting, informal orthography (elongations such as ดีมากกก, the laughter numeral 555), and, in our source domains of restaurant reviews and tweets, a culture of exuberant sincere praise that is easy to mistake for irony. If multi-agent deliberation helps anywhere, a task defined by hidden intent should be favorable ground.

Our contributions:

- (1) A human-labeled Thai sarcasm gold set with a sharp operational rubric (sarcasm requires *feigned praise*, การเสแสร้ง), documented provenance and bias notes, plus a 4.7 $\times$ -expanded hard variant labeled blind through a purpose-built tool.
- (2) A five-system, four-axis comparison (F1, calls, cost, latency) under one measurement harness, with paired bootstrap and McNemar tests throughout.
- (3) A negative result with an identified mechanism: multi-agent gains over a *confidence-aware* single agent are statistically unresolvable while costing 1.8–7.4 $\times$, because half of the apparent gain was the baseline discarding confidence it had already paid for.
- (4) Cost-axis anchors: a \$0 floor at F1 $\approx$ 0.59 reached independently by three free methods, and a confidence router giving 95% of flagship quality at 40% of its cost.

2 Task and Data

Definition. An item is sarcastic (label 1) iff the author feigns praise or thanks while intending the opposite. The rubric resolves the three boundary cases that dominate disagreement: direct complaints are 0 (negative content without pretense), balanced reviews are 0 (genuine praise plus genuine criticism), and exuberant sincere praise is 0. Undecidable items (context-dependent intent) are excluded rather than guessed.

Gold set. 72,865 texts from Wongnai restaurant reviews and the Wisersight social-media corpus were keyword-scored for sarcasm suspicion; 400 sampled items were pre-labeled by GPT-4o and then human-reviewed *blind* (the annotator decides before seeing the model's draft). Agreement analysis showed the LLM over-flags sarcasm, of 17 LLM-positive items in the first review pass, the human kept 3, motivating blind review as a design rule. Iterative mining of high-suspicion texts brought the canonical set to **127 items (30 sarcastic / 97 not)**. Provenance caveats (some positives were mined with the same model family later evaluated, likely inflating its recall) are documented and affect all systems equally.

Hard expansion. A later blind-labeling campaign exhausted the mined candidate pool (495 items), giving **gold_v2: 595 items (104 sarcastic, 17.5%)**. Because candidates were pre-selected to *look* sarcastic and 81% were judged sincere, gold_v2 is deliberately enriched with plausible negatives, a hard set whose absolute scores are not comparable to the canonical set (§6.4).

3 Systems

All LLM systems use the same detector prompt and JSON-constrained output on GPT-4o unless stated; prices are July-2026 list prices.

S1 Single agent. One call per item, argmax label. **S1+ τ Threshold.** Identical calls, but the positive-class probability is read from token logprobs and thresholded; τ is chosen leave-one-fold-out (stratified 5-fold, seed 42), so no item influences the threshold that judges it. **S2 Pipeline.** A screener flags candidates; a *constrained* verifier may only overturn 1 $\rightarrow$ 0 and only when confident, preserving the screener's recall. **S3 Debate.** Prosecutor and defender argue; a judge decides freely (3 calls/item). **S4 Hybrid.** A four-agent committee mixing the above roles. **S5 WangchanBERTa.** A Thai transformer fine-tuned on the gold set under 5-fold cross-validation $\times$ 3 seeds (every prediction from a model that never saw the item), run locally at \$0.

4 Evaluation Protocol

Accuracy is not reported: with 76% negatives, constant rejection scores 0.76. All systems predict the same items, so comparisons are paired: F1 differences carry 95% CIs from a paired bootstrap (5,000 resamples of the item set), and label flips are counted with McNemar's test. Costs are computed from measured token usage; latency is wall-clock per item (p50).

5 Results

System	F1	Prec.	Rec.	Calls	Cost	p50
S1 Single agent (argmax)	0.690	0.526	1.000	127	\$0.094	751 ms
S1+ τ Single agent + threshold	0.725	0.641	0.833	127	\$0.094	751 ms
S2 Two-agent pipeline	0.744	0.604	0.967	183	\$0.169	967 ms
S3 Debate (3 agents)	0.694	0.595	0.833	381	\$0.695	4,557 ms
S4 Hybrid (4 agents)	0.700	0.560	0.933	292	\$0.407	832 ms
S5 WangchanBERTa (local)	0.620	0.553	0.700	0	\$0.00	26 ms

TABLE 1. All systems on the same 127 items, same measurement code.

5.1 The fair baseline erases the multi-agent win

Against the plain single agent, the pipeline's +0.054 looks meaningful (95% CI $[-0.002, +0.113]$, $P(\text{not better}) = 3\%$). But that baseline throws away the model's own confidence. Reading it back, zero extra calls, zero extra dollars, lifts the single agent to 0.725, and against *that* baseline every multi-agent system is unresolved: pipeline +0.019 $[-0.073, +0.113]$ at $1.8\times$ cost; hybrid -0.025 $[-0.117, +0.071]$ at $4.3\times$; debate -0.030 $[-0.158, +0.096]$ at $7.4\times$. Roughly half of the apparent multi-agent gain was information the baseline had already paid for and discarded.

5.2 Architecture: constraints beat head-count

Why does the two-agent pipeline outperform the three-agent debate ($\Delta +0.049$, at a quarter of debate's price)? The detector starts from recall 1.000; what remains is buying precision without spending recall. The pipeline's verifier is *structurally constrained*, it can only overturn positive verdicts, and only when explicitly confident, so it keeps recall at 0.967 while pruning false positives. Debate returns full freedom to the panel: its defender successfully argues away 5 true positives (recall 0.833), spending the asset the detector had secured. The number of agents matters less than the constraint structure imposed on them.

5.3 Model choice dominates architecture choice

The tuned single agent across five OpenAI models spanning a $25\times$ price range stays within a 0.05 F1 band, with the best score from one of the cheapest (gpt-4.1-mini: F1 0.727, \$0.015). That cheap single agent is statistically tied with the flagship pipeline (gpt-4o, 0.744, \$0.169) at **1/11 of the cost**. Local open models (Qwen2.5-7B, SeaLLM-7B, Typhoon-8B, OpenThaiGPT-7B, 4-bit, chat-templated, same leave-fold-out protocol) top out at 0.576, a real but modest gap of +0.152 to the GPT bot, and Thai-specialized models show no advantage over general Qwen.

5.4 Anchoring the cost axis

The \$0 floor. Three independent free methods, the regex 555 alone (F1 0.590), an e5 embedding kNN (0.588), and fine-tuned WangchanBERTa at the default threshold (0.590), converge at ≈ 0.59 . The floor is a property of the task, not of any one method. **A useful middle.** A three-way router (confident-negative / uncertain $\rightarrow$ escalate to GPT / confident-positive) at a 40% escalation budget reaches F1 0.690, 95% of the GPT bot, for 40% of its cost; beyond that the curve is flat. A cautionary side-result: tuning WangchanBERTa's threshold leave-fold-out *underperforms* not tuning at all (0.556 vs. 0.590) at this n , at small sample sizes, a tunable knob is a liability, and the 0.071 "gain" from tuning on the full set is leakage.

5.5 The deployed cascade: a free abstention knob beats a model tier

The shipped tool escalates only cue-uncertain items to the LLM. Does inserting the fine-tuned WangchanBERTa as a free middle tier (cue $\rightarrow$ WangchanBERTa $\rightarrow$ GPT) help? Measured offline on the 127-

item set with leakage-free out-of-fold WangchanBERTa probabilities and cached gpt-4.1-mini scores, the outcome splits into a real lever and an inert one. The lever is a free *abstention* threshold on the cue tier: letting the lexical model answer only when its combined log-lift exceeds log 2.46 (one strong cue) and escalating the rest lifts F1 from 0.627 to **0.806** while paying for 42% of the calls, above GPT-on-everything (0.746) at under half its cost and stable under cross-validation (0.802 ± 0.023). The inert component is the model tier itself: replacing every WangchanBERTa decision with the constant "not sarcastic" under identical routing yields an *identical* 0.806; the middle tier's paired Δ F1 over cue→GPT is 0.000; its answers are not seed-stable (0.769 / 0.806 / 0.806 across three seeds); and the confident-negative threshold calibrated out-of-fold does not transfer, on ten unseen sentences the deployed model's sarcasm probability never falls below 0.37, so the tier fires on none of them. This is a third instance of the paper's pattern: a free knob that gates the confidence already present outperforms inserting an additional model.

5.6 Replication at 4.7× the data

System (gold_v2, n=595)	F1	Prec.	Rec.	Cost
S1 Single agent (argmax)	0.337	0.206	0.933	\$0.275
S1+τ + threshold (LOFO)	0.377	0.238	0.894	\$0.275
S2 Two-agent pipeline	0.357	0.222	0.913	\$0.814 (2.96×)

TABLE 2. Core systems on the expanded hard set. Absolute scores are not comparable to Table 1 by design (§6.4).

On the expanded set two things resolve. First, the confidence-reading upgrade that defines our fair baseline is now *statistically significant*: +0.040 F1 over the plain baseline, 95% CI [+0.018, +0.060], McNemar 80:6, at identical cost and call count. What was a suggestive trend at n=127 and n=302 is confirmed at n=595. Second, the thesis comparison holds and hardens: measured against that baseline the pipeline is **−0.019** (95% CI [−0.047, +0.009]), a 91% chance it is not better, McNemar 49:83 in the single agent's favour, at 2.96× the cost. The interval has narrowed monotonically with sample size (0.186 at n=127, 0.094 at n=302, 0.056 at n=595) without the sign of the conclusion ever flipping toward multi-agent.

Absolute F1 falls to ≈ 0.35 for every system because the added items are adversarially plausible negatives: 403 of the 495 mined candidates (81%) were judged sincere by the human annotator, diluting the positive rate to 17.5% and surging false positives identically across architectures. The model is also more overconfident at this scale (404/595 items outside [0.01, 0.99]), yet the threshold still extracts a significant gain, and its leave-fold-out score (0.377) sits within 0.004 of the oracle-tuned value (0.381), indicating a stable rather than overfitted cutoff.

5.7 Reality check on a uniformly random test set

Every set above was keyword-mined to 17--24% sarcastic. To measure the true base rate we drew 250 uniformly random texts from the 72,865-item pool, labeled blind: only **4.9%** (12/245 decidable) are sarcastic, so keyword mining inflates the positive rate roughly 4.5×. On this realistic distribution the class-imbalance mathematics dominates every result.

System (random set, n=245, 4.9% sarcastic)	F1	Prec.	Rec.	Flag rate
Always "not sarcastic"	0.000	—	0.000	0%
Lexical cues (deployed free tool)	0.090	0.055	0.250	22%
GPT-4o (argmax)	0.202	0.115	0.833	36%
GPT-4o + threshold (LOFO)	0.462	0.429	0.500	6%

TABLE 3. Absolute performance on a realistically distributed test set. Threshold F1 95% CI [0.182, 0.688] (wide: only 12 positives).

Two anchors from the mined data are exposed as artifacts. The "\$0 floor at F1 0.59" (the free cue detector) collapses to **0.09**: the same cues that fire on sarcasm in a 24%-positive set fire overwhelmingly on sincere text when positives are rare (52 false positives), so precision falls to 0.055. This is not a property of the method but the arithmetic of a rare positive class. GPT with argmax fails identically, flagging 36% of all text as sarcastic (precision 0.115) when only 4.9% is.

The one intervention that survives is the free confidence-reading upgrade that defines our fair baseline. Thresholding the logprob cuts the flag rate from 36% to 6% (near the true 4.9% base rate), lifting precision from 0.115 to 0.429 and F1 from 0.202 to **0.462**: a +0.26 gain at zero added cost, larger than any architectural change measured in this work. On realistic data, reading the confidence already paid for is not a marginal upgrade but the line between a usable and an unusable system. The best absolute F1 achievable here is ~ 0.46 , well below the 0.73 headline from the mined set; sarcasm detection on real traffic is materially harder than any mined benchmark implies.

6 Limitations

(1) **Scale**. 30 positives (canonical) cannot resolve ~ 0.02 F1 differences; the expansion raises power but is not i.i.d. (2) **Provenance bias**. Some positives were mined with the model family under evaluation; the bias is shared across systems, so paired comparisons stand, but absolute recall should be read cautiously. (3) **Domain transfer**. On 55 hand-labeled Pantip forum comments, precision falls $0.68 \rightarrow 0.40$ (sincere praise flagged as sarcasm) while recall holds; deployment outside reviews/tweets requires re-tuning on target-domain labels. (4) **Single language/task**. Conclusions are demonstrated for Thai sarcasm; the *method*, fair confidence-aware baselines, paired statistics, cost as a first-class axis, is the transferable part.

7 Conclusion

On a task chosen to favor deliberation, multi-agent LLM systems did not separate from a single agent that simply reads its own confidence, while costing $1.8\text{--}7.4\times$ more. The result replicated at double the data with threefold-tighter intervals. Before adding agents, exhaust the one you have: read its confidence, tune with leakage-free protocol, and choose the cheapest sufficient model. The same economy governs a deployed cascade's cheap tiers: a free abstention threshold on the lexical screener beat inserting a trained model between it and the LLM, which a constant-output control matched exactly. Architecture is a question to ask after the information already purchased has been spent.

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Code, data, full statistics (findings 1–19), and a live demo: github.com/Thanaphumi2006/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection · Live detector: thanaphumi2006.github.io/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection/app.html